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DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING

PhD Prospectus

**MULTI-LAYER SECURITY FOR MOBILE USERS:
MEASUREMENT AND MITIGATION FROM APPLICATIONS TO CONVERSATION**

By

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Abstract

Mobile users can lose control of their data and money through harmful applications, platform mechanisms that obscure changes in access, and deceptive conversations. This dissertation investigates these three settings through measurement and the assessment of practical defenses. Two studies provide its foundation. *The Cost of Convenience* examines loan applications across five countries and connects policy requirements to observable application behavior; coordinated disclosure led to the removal of 93 applications from Google Play. *Silent Consent, Persistent Risk* studies permission expansion and cross-application access in Android, combining a longitudinal corpus of 19.3 million APKs with targeted device experiments and a transparency prototype. The ongoing third study, *SIRENBERT*, investigates whether a sequence of observable conversational behaviors can support timely warnings about financial scams on the user's device. An encoder-recurrent-model prototype and a proxy conversation corpus provide a starting point, while annotation validation, generalization, warning timing, and deployment evaluation remain research tasks. The proposed extension examines images and other media presented as evidence of identity, circumstances, or financial opportunity. I will reconstruct conversations and embedded images from public screenshot collections, measure how these artifacts are used, and test whether their interpretation in context improves warnings over text alone. The central question is how evidence available at each layer can be translated into useful protection while preserving user privacy and controlling false alarms. The dissertation will distinguish demonstrated findings from the hypotheses that these remaining studies will test.

1 Introduction

Mobile devices mediate access to financial services, private communication, and personal information. Their importance is especially visible where mobile money provides access to services that conventional banking does not readily reach [1, 2]. This dependence creates several opportunities for abuse. An application may collect information inconsistent with its stated purpose; a platform may grant access without making the change visible; and a person using an otherwise legitimate messaging application may be persuaded to send money under false pretenses. These problems involve different evidence and intervention points, but share a concern: users may have little opportunity to recognize a developing risk before a consequential action occurs.

My dissertation examines this concern at the application, platform, and conversation levels. The first study measures how loan applications comply with regulatory and platform requirements and how their runtime behavior exposes sensitive data [3]. The second investigates Android permission mechanisms that allow capabilities or cross-application access to expand without renewed user visibility [4]. Together, these studies establish methods for connecting policy or permission abstractions to observable behavior and for assessing interventions in realistic settings.

The remaining research addresses conversational financial exploitation. Access controls can constrain an application’s use of data, but they do not determine whether a request made through that application is trustworthy. A request for money can be legitimate, and behaviors such as affection, professional self-presentation, and urgency also occur outside scams. The research problem is therefore to determine whether the evolving context of an exchange provides useful evidence for a timely warning, without treating ordinary persuasion or financial discussion as sufficient evidence of fraud.

Thesis proposition. *Measuring how access and trust develop over time can reveal risks that are obscured by isolated application, permission, or message observations. These measurements can inform defenses at the point where the relevant evidence is available, with their value assessed through detection quality, timeliness, user visibility, and deployment cost.*

This proposition does not require a single detector for all three settings. The completed studies provide evidence at the application and platform levels. The ongoing conversation study tests behavioral sequence modeling, and the proposed extension asks whether images presented within an exchange add useful evidence. Sections 3 and 4 summarize completed work; Sections 5 and 6 specify what remains to be established.

2 Background and Research Gaps

2.1 Application behavior and permission visibility

Digital lending can improve access to credit, but loan applications also create opportunities for intrusive data collection and coercive debt recovery. Registration requirements and platform permission policies provide complementary constraints. The application-level question is whether the behavior of approved applications is consistent with those requirements, rather than whether approval alone indicates safety. My loan-app study connects policy interpretation to static and dynamic measurements [3].

Android permissions have been studied in terms of application privileges and users’ understanding of access requests. Felt et al. examined permission use and user attention, while Wijesekera et al. investigated the context in which users make permission decisions [5–7]. Calciati et al. measured automatically granted permissions across application versions; Tuncay et al. studied custom-permission weaknesses; and Gamba et al. examined preinstalled software [8–10]. My platform study builds on this work by measuring permission evolution at larger scale and testing selected access paths on Android 16. Its prototype addresses visibility into changes, rather than claiming to prevent all misuse of an otherwise granted permission.

2.2 Conversational scams and evidence beyond text

Whitty describes the persuasive progression of romance fraud. Oak and Shafiq use victim interviews to examine relationship-based investment scams, including trust development, financial exploitation, and subsequent re-engagement. Acharya and Holz investigate the practices and infrastructure associated with pig-butchering scams [11–13]. These studies motivate examining connected behaviors over time; they do not imply that every scam follows one fixed sequence or that a particular tactic uniquely identifies fraud.

Recent detection work provides useful comparison points. Lu et al. study mobile messaging scams using Reddit data, Sun et al. introduce a benchmark for scam progression, and Topcuoglu et al. compare generative-model approaches to scam detection [14–16]. Their settings and data sources motivate complementary questions about message-level evidence, transfer between sources, and device constraints. Our scam-family labels will describe our own data; an external benchmark’s categories are supporting context rather than a required taxonomy.

The remaining gap is the reliability and utility of a representation that can be updated as a private conversation unfolds. This requires appropriate negative examples, validated message labels, and evaluation before the financial request. It also raises a multimodal question: a photograph, document, or balance screenshot may be presented as proof, while its relevance to a later request depends on what has already been said. Detecting image reuse or manipulation alone does not establish that an exchange is exploitative. The proposed work studies the artifact’s role and contextual interpretation together.

3 Completed Work I: The Cost of Convenience

The Cost of Convenience: Identifying, Analyzing, and Mitigating Predatory Loan Applications on Android, published at ACM ASIA CCS 2026, investigates whether loan applications comply with national rules and Google’s Financial Services Policy [3].

Problem and approach. An approved lender can still operate an application that collects prohibited data or exposes it before a borrower has completed registration. We studied 434 applications from registries and markets in Indonesia, Kenya, Nigeria, Pakistan, and the Philippines. An LLM-assisted policy-to-permission mapping translated policy text into testable checks, which we combined with static code inspection and dynamic observation of network and data-access behavior.

Findings and contribution. Among approved applications, 141 violated national regulatory requirements and 147 violated platform policy. In the dynamic analysis, 37 of 148 functioning applications transmitted sensitive information at launch after permissions were granted, before signup or a loan application. Following coordinated disclosure, Google removed 93 flagged applications, representing over 300 million cumulative installs. These installation counts indicate potential exposure, not a count of verified victims. The study contributes a measurement method and evidence for more consistent compliance assessment across regulatory and platform requirements.

4 Completed Work II: Silent Consent, Persistent Risk

Silent Consent, Persistent Risk: Android Permission Groups and Custom Permissions examines how application capabilities and data access can change without corresponding user visibility [4].

Problem and approach. A permission decision made for one application version may not convey what a later version can access. Custom permissions can also protect components in ways that admit unrelated applications. We analyzed 19,324,760 APKs representing 5,965,583 applications in AndroZoo [17], tracked permission changes, and compared their association with VirusTotal detections across thresholds [18]. Targeted Android 16 experiments tested whether selected mechanisms remained exploitable. A prototype using

public Android APIs exposed otherwise silent permission changes to the user.

Findings and limits. Of 2,244,575 multi-version applications, 381,026 (17%) exhibited within-group permission expansion. Malware-flagged applications had higher odds of expansion at the primary threshold (odds ratio 1.35), an association rather than evidence that every expansion is malicious. Static analysis identified 307 active cross-developer access pairs. Dynamic retrieval of sensitive data was confirmed for the 39 pairs amenable to probing on Android 16; the dynamic claim does not extend to all 307 pairs. In a 96-day single-device pilot, the transparency prototype reported 23 expansion events across 13 applications. This establishes feasibility of providing visibility, while broader usability and protective effects require further evaluation.

5 Ongoing Work III: Conversational Scam Detection

SIRENBERT investigates whether a mobile device can recognize a developing conversational scam from the evidence available at each point in an exchange. The work completed so far comprises a problem definition, a proxy corpus, a scam-family mapping, a behavioral representation, preliminary measurements, baseline comparisons, an encoder–recurrent-model prototype, and an Android integration. The revised annotation and model design described below is the current specification; its large-scale validation, retraining, generalization tests, and warning evaluation remain dissertation work.

5.1 Problem definition and observable decision rule

A *conversational scam* is a multi-turn exchange in which a suspect, under false pretenses, induces a target to part with money or other economic value through a request or instruction. This definition covers romance, investment, advance-fee, employment, marketplace, impersonation, technical-support, and related fraud when the financial action is requested through the exchange. The current evaluated scope requires at least four observed messages and participation by both parties. It excludes isolated solicitations and coercion-only sextortion, which do not require the conversational development studied here.

The broad definition describes the phenomenon, but a transcript cannot establish whether every claimed identity, emergency, or promised return is false in the world. The operational rule therefore uses observable evidence. A suspect message is a *financial demand* when it asks or instructs the target to take an action that would cost money or other economic value. A demand becomes *tactic-linked* when an earlier or co-occurring tactic in the observed exchange is explicitly connected to that demand. No completed payment is required. A demand without such evidence is recorded as *not yet tactic-linked*; this describes what is observable at that point and does not assert that the request is legitimate.

Let $\mathcal{B}_t$ be the set of behaviors assigned to suspect message t , let D denote `FinancialDemand`, let $\mathcal{T}$ be the set of tactic triggers, and let $R(j, t) = 1$ when reviewed evidence connects tactic j to demand t . The causal status of a demand is

$$\ell_t = \mathbb{I}[D \in \mathcal{B}_t] \mathbb{I}[\exists j \leq t : \mathcal{B}_j \cap \mathcal{T} \neq \emptyset \wedge R(j, t) = 1]. \quad (1)$$

Here $\mathbb{I}[\cdot]$ is an indicator: it equals one when the condition in brackets is true and zero otherwise. Consequently, $\ell_t = 1$ only for an observed demand with at least one reviewed tactic connection available by time t . The conversation label is binary:

$$Y(C) = \begin{cases} \text{SCAM}, & \exists t : \ell_t = 1, \\ \text{NON-SCAM}, & \text{otherwise.} \end{cases} \quad U(C) = \mathbb{I}[\exists t : \mathcal{B}_t \cap \mathcal{T} \neq \emptyset]. \quad (2)$$

Here $U(C)$ records whether a scam-like trajectory is present; it is not a third label. Thus a conversation

containing tactics but no linked demand remains NON-SCAM with $U(C) = 1$. If a first demand is not tactic-linked, later tactics can still support a warning before a repeated demand; later evidence does not retroactively change what was observable at the first demand.

The project uses each eligible transcript as observed. A screenshot may begin mid-conversation, and a complete transcript need not begin with a greeting. Extraction quality therefore checks whether visible messages were recovered in order with correct speakers, while capture format is retained as metadata for stratified analysis. Neither a missing greeting nor source provenance determines the operational label.

5.2 Proxy corpus and scam-family coverage

Complete private scam conversations are scarce and difficult to release. We assembled a proxy corpus from recorded scam exchanges, public scambait screenshots, ordinary dialogue, legitimate persuasive requests, and generated conversations. LoveFraud02 contains researcher-recorded exchanges with purported romance-fraud perpetrators, while Reddit Scambait contains screenshot-based encounters in which people knowingly engage suspected scammers [14, 19]. These sources provide real scam language but are not a representative sample of unsuspecting victims: scambaiting is performative, screenshot sequences can omit context, and public reports are selected for posting.

Topical-Chat and XDailyDialog provide ordinary dialogue, and PersuasionForGood provides legitimate requests for charitable donations [20–22]. The latter supplies a difficult comparison because persuasion and a financial request are present without fraud. Generated pig-butcher and ScamForge conversations extend combinations that are rare in the real collections. Their generation condition is retained as provenance rather than treated as a validated verdict, and results on real and generated sources will be reported separately.

A separate live and archival Reddit collection has indexed 92,595 distinct scambait posts and 319,008 Reddit image attachments, most of which are chat screenshots. The latter is a media-index count that includes unavailable files; 251,624 entries were on disk or recovered at the latest completed audit. A screenshot may contain multiple visible chat messages, and a post may contain multiple screenshots, so neither figure represents extracted conversations or messages. Series reconstruction preserves the evidence used to join posts that appear to belong to one exchange. These records are not included in Table 1: screenshot extraction, speaker and message-order checks, deduplication, and a held-out quality audit must precede their admission to the conversation corpus. The same image-preserving collection also supplies the primary candidate pool for the proposed multimodal study in Section 6. It retains the source images needed to recover both message text and embedded credibility artifacts such as identity documents, receipts, photographs, and balance or investment screenshots.

To describe coverage without conflating a scam name with its mechanism, the current mapping follows

Source	Conversations	Suspect messages
LoveFraud02	83	26,423
Reddit Scambait	1,166	65,685
Topical-Chat	8,628	97,204
XDailyDialog	3,100	19,981
PersuasionForGood	1,017	10,600
Generated pig-butcher scenarios	5,683	237,296
Generated ScamForge scenarios	3,097	152,371
Total	22,774	609,560

Table 1: Assembled proxy corpus. Counts describe suspect-side messages available to the annotation pipeline. Source groups describe how conversations were obtained; they are not the final transcript-derived labels.

seven expected-benefit or expected-consequence families from Beals et al. [23]. Each conversation receives one primary family; relationship use, impersonated identity, payment mechanism, and a common name are orthogonal modifiers. For example, pig butchering that promises returns belongs to Consumer Investment, with `relationship_used=true` and `named_scam=pig_butchering`. “Advance fee” records how value is taken and can occur in several families, so it is not a universal primary family.

An initial model-assisted pass over 520 in-scope real-source candidates proposed 218 Relationship/Trust, 78 Products/Services, 53 Prize/Grant, 31 Investment, 13 Employment, eight Charity, and two Phantom Debt cases; it also returned 115 NONE and two unparsed outputs. That distribution is provisional. An automated audit flagged 67 of the 520 outputs. Independent review of the 115 NONE cases agreed on the apparent family in 97 cases and assigned 108 to one of the seven families, showing that the unreviewed model output substantially undercounted several families. The final distribution will be reported only after all flagged and probability-sampled cases receive a second reading.

5.3 Lifecycle groups, triggers, and forms

The representation separates a conversation family from what each message does. It uses four non-exclusive evidence groups:

$$\text{Contact} \longrightarrow \text{Relationship} \longrightarrow \text{Tactics} \rightleftharpoons \text{Financial Demand}. \quad (3)$$

The arrows summarize common progressions rather than impose a required order. Groups may be skipped, repeated, or revisited after a demand. In particular, Relationship is not a benign state, Tactics is one group rather than three states, and Financial Demand is an observable act rather than an automatic verdict.

The four groups are separated by the function a behavior serves in relation to financial extraction, as summarized in Table 3.

The central boundary is between Relationship and Tactics. We apply a functional test: could the observed behavior serve as the reason, condition, or lever for a payment request? Routine conversation, personal disclosure, and affection build the interaction but do not, by themselves, supply that basis. A claimed identity or status, promised return, crisis, verification avoidance, request for secrecy, deadline, or emotional condition can do so. This division parallels prior lifecycle accounts that distinguish grooming from the sting and bonding from the bait or squeeze [11, 12].

Romantic affection therefore remains a Relationship behavior even when it occurs inside a romance scam. For example, “I love you” does not itself make a later request for money tactic-linked. A fabricated hospital crisis, or making affection conditional on payment, supplies Tactics evidence that may support such a link. Operationally, Relationship evidence alone cannot satisfy the demand-link relation in Equation 1. Tactics evidence is eligible to do so only when the annotation records an explicit connection to the demand. A demand for which no such connection is yet visible is `not_yet_tactic_linked`; this describes the observed evidence and does not establish that the request is legitimate.

The candidate annotation vocabulary contains thirteen observable triggers (Table 4). Every suspect message receives at least one trigger, and independently supported triggers can co-occur. *Routine Engagement* labels a greeting, acknowledgment, check-in, joke, or everyday exchange that keeps an established conversation active when no more specific behavior is expressed. It is therefore a positive account of ordinary maintenance rather than an “other” category. The first visible message is assigned *Contact Initiation* only when its content or visible context actually establishes the interaction.

This granularity follows a functional rule: a top-level trigger must change what the sequence model should retain about the interaction. Closely related content with the same function remains a form. Thus a claim of authority and a claim of wealth are distinct Credential Positioning forms because both attempt to raise the standing on which later claims are believed; the planned ablation will test whether retaining those

Primary family	Organizing expectation	Examples or modifiers
Consumer Investment	Return on an investment	crypto, forex, investment-return pig butchering
Consumer Products/Services	Product, service, credit, or sale	marketplace, fake buyer, tech support, recovery, fake loan, paid service
Employment	Income from work	fake job, reshipping, mystery shopper, multilevel marketing
Prize/Grant	Prize, grant, inheritance, or windfall	lottery, sweepstakes, inheritance, government grant, Nigerian letter
Phantom Debt Collection	Avoiding a claimed debt consequence	tax, toll, court, or medical debt
Charity	Supporting a claimed cause or hardship	disaster, religious, personal appeal
Relationship/Trust	Maintaining or helping a relationship	romance, allowance, grandparent, friend/relative, boss request

Table 2: Conversation-level family mapping. These labels describe corpus coverage and support family-held-out evaluation; the completed-conversation family label is not supplied to the causal detector.

Group	Function in the observed conversation
Contact	Opens the interaction.
Relationship	Builds or maintains the interaction itself.
Tactics	Creates a reason, condition, or pressure for the target to give up economic value.
Financial Demand	Asks the target to give up that value.

Table 3: Functional definition of the four lifecycle groups. The groups describe what a behavior does, rather than require it to occur at a fixed point in time.

forms improves detection. The inventory claims coverage of observable behaviors relevant to the in-scope scam definition, not every act in human conversation. Its coverage will be tested by residual review across all seven families, explicit searches for missed acts, co-occurrence analysis, and inter-annotator agreement. A new trigger is added only when reviewed examples show an independently observable function that the existing triggers cannot represent and that Stage 2 can use.

For interpretation, define a four-dimensional accumulated-presence summary

$$\mathbf{s}_t = [c_t, r_t, u_t, d_t], \quad \mathbf{s}_t = \mathbf{s}_{t-1} \vee G(\mathcal{B}_t), \quad (4)$$

where G maps the current behavior set to Contact, Relationship, Tactics, and Financial Demand, and $\vee$ is element-wise logical OR. This summary permits skipped groups but loses repetitions after a bit first turns on. The learned model therefore receives the full ordered per-message sequence; Equation 4 is used only for explanation or as an ablation.

For a financial demand, annotation records only information that has a defined role: one or more payment methods and the evidence links used to validate Equation 1. The closed payment list is cash, bank or wire, card, gift card, cryptocurrency, peer-to-peer application, cheque, online platform, mobile money or airtime, goods or property, and unspecified. Requested amount and recipient are not mandatory structured fields: an exploratory scan found amount mentions at similar rates in scam and benign messages, while those values can still be extracted later for descriptive analysis.

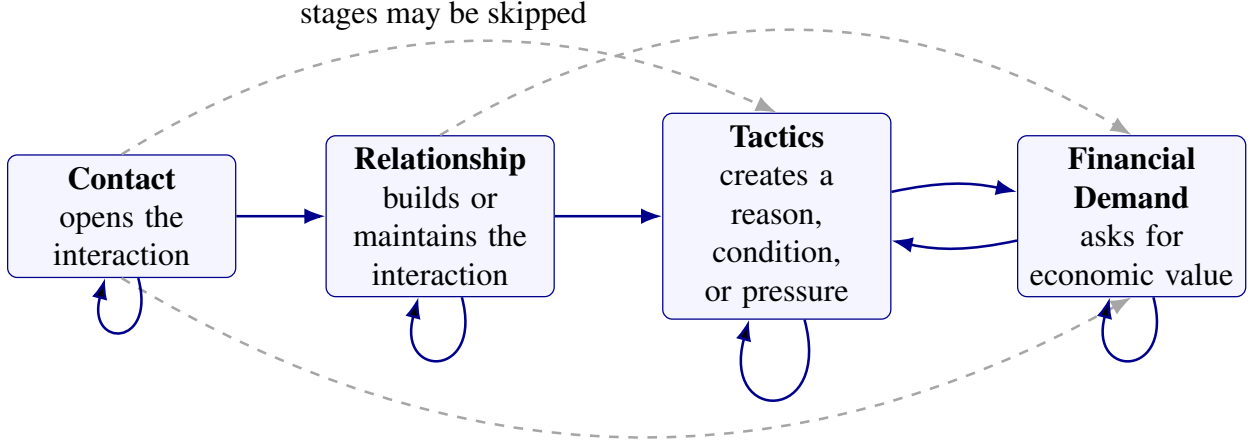


Figure 1: Current lifecycle representation. The detector consumes the ordered message sequence; the four groups organize evidence for interpretation and do not force a monotone state path.

5.4 Preliminary measurements and baselines

The earlier single-label annotations provide motivation, not final ground truth. Under their automated rule, 520 real-source conversations contained a qualifying demand. They had a median of two distinct mapped indicators before the first demand, and 52.3% had at least two. Among 1,685 Topical-Chat, XDailyDialog, and PersuasionForGood conversations containing a demand but not receiving that scam label, 90.1% contained no mapped indicator before the demand. Indicator frequency also varied with length: a money-making offer appeared in 33.6% of the shortest quartile but 13.2% of the longest, while affection and emotional leverage became more common in longer conversations. This is consistent with shorter investment approaches and longer relationship approaches, and cautions against treating one tactic as necessary.

A conversation-level audit selected one demand from each of 751 rule-positive and 1,337 rule-negative conversations in the real-source subset. The first selected demand occurred within the first four suspect messages in 53 of 751 positives (7.1%) and 735 of 1,337 negatives (55.0%). Gift-card, cryptocurrency, or peer-to-peer payment language appeared in 98 positives and eight negatives; among the early demands it appeared in nine positives and five negatives. These values show why payment method is a candidate feature, but they do not estimate field precision. The benign sources contain little realistic person-to-person payment conversation, so payment-method gains must be tested on a dedicated benign payment stress set and under source-held-out evaluation.

We also compared demand-only and completed-conversation baselines over all 22,774 conversations. For this preliminary comparison only, source provenance supplies the positive and negative groups because the revised transcript labels are not yet validated. The five encoders were trained previously to recognize heuristic financial-demand labels; the zero-shot Gemma 4 E4B judge received the completed conversation. Table 5 therefore measures separation of the assembled source collections rather than performance against the final operational rule.

The demand detectors cluster between $F_1 = .705$ and $.718$, supporting the motivation that recognizing a request for money does not determine whether it is fraudulent. The completed-conversation judge reaches $F_1 = .896$ but still misses 296 scam-source conversations and flags 797 benign-source conversations; it also has access to future messages unavailable to a causal detector.

The implemented encoder-GRU prototype supplies a second starting point. Nested five-fold cross-validation selected ModernBERT with ten preceding messages. On the original 1,297-conversation test

Group	Trigger	Observable function	Closed forms, when used
Contact	Contact_Initiation	Attempts to establish the interaction	ordinary greeting; wrong number; mistaken identity; direct approach
Relationship	Routine_Engagement	Keeps an established interaction active	—
	Personal_Disclosure	Shares or elicits personal, vulnerable, or intimate information	—
	Affectionate_Bonding	Builds or intensifies affection or romantic commitment	warmth/endearment; romantic interest; commitment
Tactics	Credential_Positioning	Invokes identity, standing, resources, expertise, access, or connections for credibility	identity/relationship; authority/status; wealth/resources; profession/expertise; access/connection
	Gain_Offer	Presents an economic opportunity or benefit	investment return; employment income; prize/grant/inheritance; loan/credit/debt relief; gift/allowance; product/service; donation/support
	Crisis_Disclosure	Presents an emergency or acute hardship	—
	Verification_Avoidance	Evades an independent check or substitutes controlled proof	—
	Platform_Migration	Requests movement to another communication channel	—
	Secrecy_Request	Requests concealment or deletion of evidence	—
	Urgency_Pressure	Uses a deadline, scarcity, or demand for immediate action	—
	Emotional_Leverage	Makes affection, guilt, obligation, approval, fear, or relational loss conditional on compliance	—
Demand	Financial_Demand	Requests or instructs an action costing economic value	payment method recorded separately

Table 4: Candidate multilabel trigger vocabulary and closed forms. Forms preserve distinctions within a shared conversational function without turning each surface realization into another top-level trigger.

partition and its original labels, it obtained scam $F_1 = .863 \pm .005$ across seeds 7, 42, and 123, recall from .875 to .917, and eight to ten false positives among 1,249 original negatives. Per-seed bootstrap intervals spanned [.775, .933]. Re-scoring the unchanged models against the provisional revised labels reduced mean F_1 to $.701 \pm .007$ and mean recall to .610 because the positive count changed from 48 to 70. This is evidence of label mismatch, not a retrained result; the revised annotation must be validated before the model is retrained and a new headline value is reported.

5.5 Revised two-stage model

Stage 1: behavioral interpretation. Let $x_{1:t}$ denote the current message and all context from both speakers available through that message. DistilBERT, ModernBERT, and an adapted on-device language model will be compared as contextual encoders [24–26]. Each encoder produces a shared representation

$$\mathbf{h}_t^{\text{enc}} = \text{Encoder}_{\theta}(x_{1:t}). \quad (5)$$

Thirteen independent sigmoid outputs preserve co-occurring behaviors:

$$\mathbf{p}_t^{\text{trigger}} = \sigma(\mathbf{W}_{\text{trig}} \mathbf{h}_t^{\text{enc}} + \mathbf{b}_{\text{trig}}) \in [0, 1]^{13}. \quad (6)$$

These values need not sum to one. A softmax would force one label and reproduce the loss of information observed when an urgent gift-card demand could be labeled only as Financial Demand.

Four conditional form heads represent four Contact forms, three Affectionate Bonding forms, five Cre-

System	Scam F_1 [95% CI]	Scam-source missed	Benign-source flagged
Keyword financial-demand rule	.718 [.709, .727]	797	2,529
DistilBERT demand classifier	.710 [.701, .719]	789	2,676
BERT-base demand classifier	.712 [.703, .721]	791	2,634
FinBERT demand classifier	.713 [.703, .722]	795	2,620
DeBERTa-v3 demand classifier	.705 [.695, .714]	842	2,667
MobileBERT demand classifier	.716 [.706, .725]	808	2,546
Zero-shot completed-conversation judge	.896 [.890, .903]	296	797

Table 5: Preliminary source-proxy baselines. Confidence intervals use 5,000 multinomial bootstrap draws. The completed-conversation judge can use evidence after a demand and therefore is not an early-warning baseline.

dential Positioning forms, and seven Gain Offer forms. For parent trigger r ,

$$\mathbf{q}_t^{\text{form}(r)} = \text{softmax}(\mathbf{W}_r \mathbf{h}_t^{\text{enc}} + \mathbf{b}_r), \quad \mathbf{p}_t^{\text{form}(r)} = p_t(r) \mathbf{q}_t^{\text{form}(r)}. \quad (7)$$

The four gated blocks concatenate to $\mathbf{p}_t^{\text{form}} \in [0, 1]^{19}$; each form loss is evaluated only when its parent trigger is present. Payment methods are multilabel because one demand can name more than one mechanism:

$$\mathbf{q}_t^{\text{pay}} = \sigma(\mathbf{W}_{\text{pay}} \mathbf{h}_t^{\text{enc}} + \mathbf{b}_{\text{pay}}), \quad \mathbf{p}_t^{\text{pay}} = p_t(D) \mathbf{q}_t^{\text{pay}} \in [0, 1]^{11}. \quad (8)$$

The value `unspecified` is positive only when a demand names none of the ten explicit methods. Payment loss is evaluated only for demand messages.

An optional learned projection retains contextual wording outside the structured inventory:

$$\mathbf{e}_t = \mathbf{W}_{\text{proj}} \mathbf{h}_t^{\text{enc}} + \mathbf{b}_{\text{proj}} \in \mathbb{R}^d, \quad \mathbf{z}_t = [\mathbf{p}_t^{\text{trigger}}; \mathbf{p}_t^{\text{form}}; \mathbf{p}_t^{\text{pay}}; \mathbf{e}_t] \in \mathbb{R}^{43+d}. \quad (9)$$

Stage 2 receives this one concatenated numeric vector, not raw text and not gold annotations. With $d = 64$, for example, $\mathbf{z}_t$ has 107 dimensions.

Stage 2: causal aggregation. A unidirectional GRU processes the ordered Stage 1 vectors [27]:

$$\mathbf{g}_t = \text{GRU}_\phi(\mathbf{z}_t, \mathbf{g}_{t-1}), \quad \hat{\mathbf{y}}_t = \sigma(\mathbf{w}^\top \mathbf{g}_t + b). \quad (10)$$

The recurrent memory can represent order, repetition, co-occurrence, demand position, and tactics that appear between repeated demands. The final family, demand links, demand-link status, and conversation label are supervision or evaluation outputs and are never detector inputs.

The design will be tested by four nested ablations: triggers alone; triggers plus forms; triggers, forms, and payment method; and the full vector with the compact text projection. The payment block will be retained only if gains survive testing on realistic benign payment conversations and source-held-out splits. The text projection can also encode source style, so it will be retained only if its gains survive both source-held-out and family-held-out evaluation. A random-split gain alone is insufficient.

5.6 Validation and remaining work

The next annotation pilot will compare the earlier single-label output with the candidate multilabel representation on a coverage-based sample containing positive and boundary cases for every trigger, form, payment method, and important co-occurrence, across transcript and screenshot sources. At least two independent annotators will label the same conversations. Validation will report prevalence, positive agreement, and a chance-corrected agreement statistic for each trigger; kappa alone is insufficient for rare labels. Reviewers will also search for relevant behaviors outside the inventory. Only after semantic disagreements and

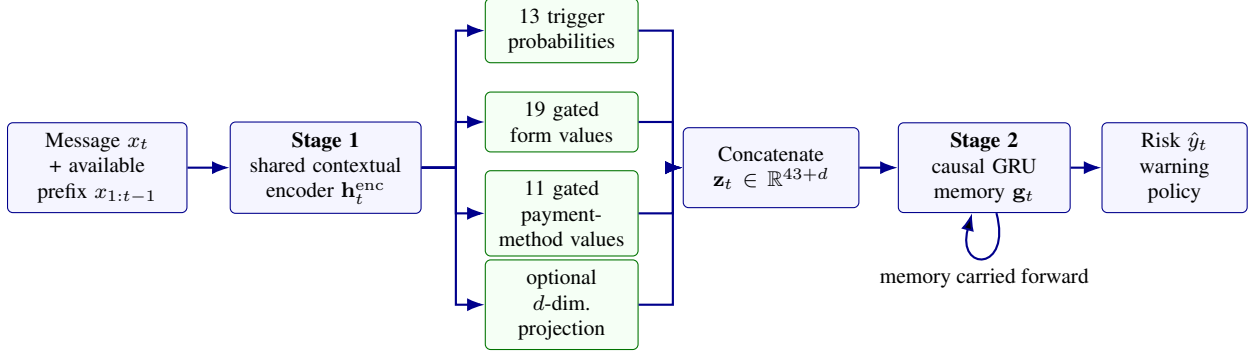


Figure 2: Revised two-stage architecture. Stage 1 predicts behaviors and selected attributes from the available prefix. Stage 2 aggregates only causal numeric outputs and produces an updated risk score after each message.

evidence-link errors are resolved with the full corpus be annotated.

RQ-T: Can a causal behavioral sequence support timely and transferable scam warnings? Training, validation, and test partitions will keep related posts, fragments, and generated variants together [28, 29]. Evaluation will report message-level trigger and attribute accuracy, conversation-level precision and recall, macro- F_1 , calibration, and false alerts per conversation. Entire-source and leave-one-family-out partitions will test whether the model learns the intended behavior sequence rather than source style or one scam family.

Warning time will be measured causally at four checkpoints: before the first observed demand, at that demand, after it but before a repeated demand, and by the end of the observed exchange. If d_1 is the first tactic-linked demand and τ is selected on validation data, define

$$\hat{t}_\tau = \min\{t : \hat{y}_t \geq \tau\}, \quad L_\tau = d_1 - \hat{t}_\tau. \quad (11)$$

Positive L_τ is a warning before the demand. Missed warnings will be reported separately rather than omitted from average lead time. At a detected financial demand below the scam threshold, the product can present a neutral pause-and-verify notice without claiming that the request is fraudulent.

Adversarial evaluation will insert plausible ordinary messages before or between tactics, because padding can push evidence outside a bounded context or dilute a learned sequence. Paraphrased tactics, delayed and repeated demands, changed payment methods, and removed platform-specific terms will test related failure modes. On-device replay will measure agreement with the research model, quantization effects, latency, peak memory, and energy on named hardware. The Android integration establishes an implementation path; the annotation study, retraining, holdout experiments, early-warning evaluation, and device measurements are the work required to establish the proposed system.

6 Proposed Work: Credibility Artifacts in Conversational Scams

6.1 Problem and motivation

A conversational claim may be supported by a photograph, identity document, receipt, or investment screenshot. I use *credibility artifact* to mean media presented within an exchange as evidence for an identity, circumstance, capability, transaction, or opportunity. The term describes its use; it does not presume that the artifact is forged or malicious. Ordinary photographs and documents may serve the same function in legitimate exchanges.

The unresolved problem is whether these artifacts provide reliable additional evidence for scam detection, and whether that evidence arrives early enough to improve warnings. A text-only model may miss what a screenshot claims; an image classifier may miss why that claim matters to the request. Conversely, an apparently inconsistent or reused image may be benign, and a scammer can use genuine media. The proposed study will measure these cases before deciding how to incorporate artifact evidence into the detector. Its primary empirical scope is images and screenshots. Audio will be a conditional extension if usable original recordings become available; an audio-message icon in a screenshot is not an audio recording.

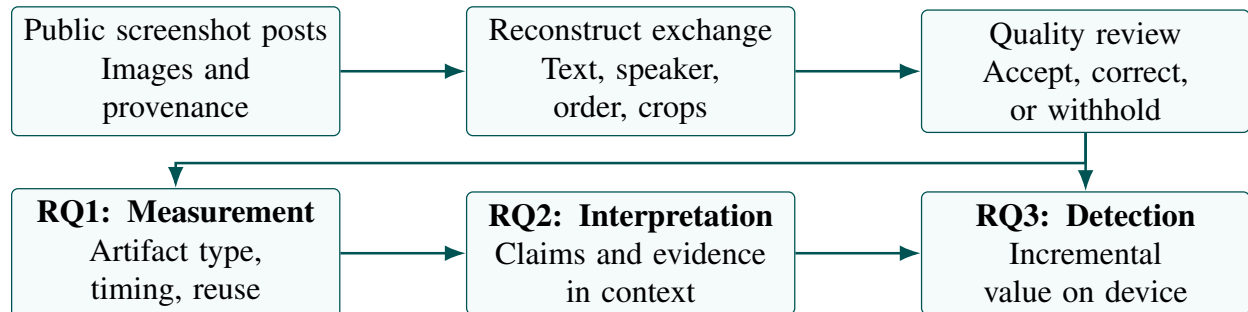


Figure 3: Proposed artifact study. Collection and extraction supply candidate evidence; quality review precedes measurement and detector evaluation. Artifact presence and source membership do not supply automatic scam labels.

RQ1: What artifacts appear, and how are they used over a conversation? Why this matters. A detector design needs evidence about which media are available, what claims they support, and when they appear. Without this measurement, a multimodal extension could focus on visually distinctive but rare or late-arriving content.

Approach. I will use the existing screenshot collection and the completed live and historical Reddit post index described in Section 5. Its full-resolution images support both conversation-text reconstruction and the recovery of media embedded within those conversations. The collection retains post identifiers, gallery order, available media, and retrieval status; media backfill and conversation extraction remain in progress. I will reconstruct messages and embedded artifact regions while retaining links to their source pixels. Text recognition, spatial layout, and a vision language model will be compared for transcription and speaker attribution. Different extraction systems’ agreement will prioritize review, but a human sample of agreeing outputs is also necessary to detect shared mistakes.

Posts will be joined into a conversation only when continuity evidence supports the connection; matching an author or title is not sufficient. Screenshot overlap will be removed without deleting genuinely repeated messages. The review will distinguish the screenshot owner’s messages from those of the suspected scammer, retain uncertain roles, and record missing sections. I will audit extraction quality by app interface, image length, compression, and source before measuring artifact prevalence, first appearance, and within/across-conversation reuse. Exact and perceptual image matches will identify reuse candidates, followed by review. A match demonstrates shared content, not identity theft or common criminal control. Synthetic-image detectors may provide exploratory scores, but those scores will not be reported as the prevalence of AI-generated scam imagery without suitable ground truth and calibration.

Evaluation and expected contribution. The study will report message coverage, transcription error, speaker accuracy, artifact crop/link accuracy, and extraction yield before giving corpus-level measurements. A held-out audit will quantify selection and reconstruction limitations. The contribution will be a provenance-preserving measurement corpus and evidence about artifact use in observed exchanges, not a claim to represent all conversational fraud.

RQ2: Does an artifact support or conflict with the claims made in context? Why this matters. Recognizing a face or extracting a balance does not establish what the sender asserts. A uniformed photograph is inconsistent with an occupation claim only if the sender presents the depicted person as themselves and the context makes those claims incompatible. The study must distinguish contradiction from irrelevance, ambiguity, and missing evidence.

Approach and evaluation. I will annotate the artifact’s visible content, the claim explicitly associated with it, and the earlier messages needed to interpret that claim. Candidate relations will include support, contradiction, and insufficient information. Text-only, artifact-only, and joint models will be compared on reviewed examples, including genuine documents, benign image reuse, edited screenshots, and scams using apparently consistent media. Controlled mismatches will test specific capabilities but will be reported separately from naturally occurring cases. Evidence localization and claim-relation accuracy will accompany classification metrics. The expected contribution is an evaluated contextual-evidence representation, including its failure and abstention cases, rather than automatic verification of identity.

RQ3: Does artifact evidence improve warnings under mobile constraints? Hypothesis. When text alone leaves a claim ambiguous, an artifact’s content and its relation to prior messages may improve detection at an acceptable false-alarm rate. Additional model complexity is justified only if the measured benefit survives source separation and device constraints.

I will compare simple score combination with learned fusion. Let $\mathbf{e}_t$ represent the artifact evidence available at message t , and let u_t indicate whether an artifact is available. A candidate extension of Stage 2 is

$$\mathbf{h}_t^{\text{multi}} = g_\psi\left(\mathbf{h}_{t-1}^{\text{multi}}, [\mathbf{z}_t; \mathbf{e}_t; u_t]\right). \quad (12)$$

When no artifact is present, $u_t = 0$ and a masked representation supplies no invented evidence. This formulation does not require turning every image type into a new trigger. The experiment will determine whether the representation and fusion mechanism help beyond the existing behavioral sequence.

Evaluation will compare text alone, artifacts alone, and the combined system on the same eligible conversations, with component ablations. Related conversations and reused-image clusters will be separated across evaluation partitions to limit memorization. I will report recall at matched false-alarm rates, warning lead time, calibration, and robustness to cropping, compression, missing media, and benign media inserted to distract the detector. Device evaluation will include latency, energy, memory, and the cost of retaining context. The default deployment design will process private media locally. Any remote lookup would require its own privacy and utility analysis; privacy does not follow from using an image hash.

6.2 Feasibility, risks, and completion criteria

A live and archival Reddit collection is complete, while media backfill and the screenshot-extraction prototype remain underway. Initial extraction runs demonstrate that structured text and artifact candidates can be recovered, while tall screenshots, speaker confusion, and tile overlap remain concrete challenges. These are feasibility observations, not a completed accuracy evaluation. Screenshot crops may be too small or degraded for reliable artifact analysis; the study will report that yield and can still characterize observable use and timing without claiming to determine authenticity.

The first milestone is a reviewed reconstruction sample and a frozen extraction contract. The second is a corpus measurement with documented selection and error rates. The third is a held-out comparison establishing when artifact evidence helps, harms, or adds no measurable value. A negative detection result remains informative if the evaluation identifies why the evidence is unavailable, unreliable, or redundant. Before release, personal identifiers will be reviewed and redacted where appropriate; data availability will reflect source permissions and human-subjects requirements rather than assume all collected media can be redistributed.

7 Research Plan and Expected Contributions

The remaining plan prioritizes reliable evidence before new model complexity. The first twelve months focus on the conversation-label pilot, reconstruction quality, text-model evaluation, and artifact measurement. The timetable below is provisional and depends on review capacity and the yield of usable media.

Period	Deliverable and decision point
Fall 2026	Prospectus defense; freeze and review the annotation guide; validate reconstruction on held-out screenshots; establish an eligible corpus and leakage-aware splits.
Spring 2027	Retrain/evaluate text models; measure warning timing, source/family transfer, and padding robustness; complete initial artifact measurements.
Summer 2027	Complete the contextual-claim reference set and compare interpretation methods; decide which artifact evidence merits detector integration.
Fall 2027	Evaluate fusion and device costs; investigate warning presentation and cross-app continuity within approved study scope. Prepare manuscripts.
Spring 2028	Resolve remaining evaluations, document limitations and artifacts, and complete dissertation writing and defense.

Table 6: Provisional milestones. Model expansion follows evidence and quality checks rather than a predetermined accuracy or publication outcome.

The completed studies contribute methods and findings about application compliance and permission visibility. The ongoing study is expected to contribute a validated behavioral representation and an evaluation of sequential, on-device scam warnings. The proposed artifact study will add a contextual measurement resource and evidence about the value and limits of multimodal detection. Together, these contributions will connect mobile-security measurements to interventions whose benefits and limitations are explicitly tested.

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